

ITA 0611 - MACHINE LEARNING

CO 1 - AT1

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COI - ATI - APPLICATION BASED QUESTIONS

A factory got three boxes containing colored balls. Box A contains 6 red, 4 blue, and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, 5 green. One box is selected at random, but Box B is twice as likely to be chosen. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C?

Given,

→ Box A: 6 Red, 4 Blue, 2 Green → 12 balls

→ Box B: 5 Red, 7 Blue, 3 Green → 15 balls

→ Box C: 8 Red, 2 Blue, 5 Green → 15 balls.

→ Box B is twice as likely as A and C.

Therefore:

$$P(A) = \frac{1}{4}; P(B) = \frac{1}{2}; P(C) = \frac{1}{4}$$

Probability of blue:

$$P(\text{Blue} | A) = \frac{4}{12} = \frac{1}{3}$$

$$P(\text{Blue} | B) = \frac{7}{15}$$

$$P(\text{Blue} | C) = \frac{2}{15}$$

Using Bayes' theorem:

$$\begin{aligned} P(C|Blue) &= \frac{P(Blue|C) P(C)}{P(Blue|A) P(A) + P(Blue|B) P(B) + P(Blue|C) P(C)} \\ &= \frac{\frac{2}{15} \times \frac{1}{4}}{\frac{1}{3} \times \frac{1}{4} + \frac{7}{15} \times \frac{1}{2} + \frac{2}{15} \times \frac{1}{4}} \\ &= \frac{2}{25} \approx 0.08 \end{aligned}$$

Answer: 8:1.

2. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is probability

Given,

Section A: 5 Red, 3 Blue, 2 Black $\rightarrow$ 10 cars

Section B: 4 Red, 6 Blue, 5 Black $\rightarrow$ 15 cars

Section C: 7 Red, 2 Blue, 6 Black $\rightarrow$ 15 cars

Section B is twice as likely,

Therefore:

Blue prob

Using

Therefore:

$$P(A) = \frac{1}{4}, \quad P(B) = \frac{1}{2}, \quad P(C) = \frac{1}{4}$$

Blue probabilities:

$$P(\text{Blue} | A) = 3/10$$

$$P(\text{Blue} | B) = 6/15 = 2/5$$

$$P(\text{Blue} | C) = 2/15$$

Using Bayes' theorem:

$$\begin{aligned} P(C | \text{Blue}) &= \frac{2/15 \times 1/4}{3/10 \times 1/4 + 2/5 \times 1/2 + 2/15 \times 1/4} \\ &= \frac{2}{21.5} \approx 0.093 \end{aligned}$$

Answer: 9.3%.

In a real-world medical image classification task for detecting breast cancer, a DL model integrated with the k-nearest consist of 1800 examples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210 True positives, determine the missing false negatives and specificity of the classifier explaining its effectiveness in medical diagnosis.

Confusion matrix :

The dataset has :

→ 1800 samples per class

→ Total = 3600

→ Testing data = 20%.

$$\text{Test Samples} = 3600 \times 20\% = 720$$

Given ;

$$TP = 210$$

$$TN = 190$$

$$FP = 30$$

Therefore :

$$FN = 720 - (210 + 190 + 30)$$

$$FN = 290$$

Accuracy :

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{210 + 190}{720} = 55.56\%$$

Precision :

$$\begin{aligned}\text{Precision} &= \frac{TP}{TP + FP} \\ &= \frac{210}{210 + 30} = 87.5\%\end{aligned}$$

Sensitivity / Recall

$$\begin{aligned}\text{Sensitivity} &= \frac{TP}{TP + FN} \\ &= \frac{210}{210 + 290} = 42\%\end{aligned}$$

Specificity:

$$\begin{aligned}&= \frac{TN}{TN + FP} \\ &= \frac{190}{190 + 30} = 86.36\%\end{aligned}$$

4. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs:

- a) Epoch 1: $TL = 2.8$, $VL = 2.4$
- b) Epoch 2: $TL = 2.5$, $VL = 2.2$
- c) Epoch 3: $TL = 2.2$, $VL = 2.0$
- d) Epoch 4: $TL = 2.0$, $VL = 2.1$
- e) Epoch 5: $TL = 1.8$, $VL = 2.3$
- f) Epoch 6: $TL = 1.6$, $VL = 2.6$

g) Epoch 7: $TL = 1.5$, $VL = 2.9$

h) Epoch 8: $TL = 1.3$, $VL = 3.2$

i) Epoch at which overfitting begins:

Overfitting begins at Epoch 4.

ii) Two techniques to reduce overfitting

Dropout - randomly disable some neurons during training, reducing dependence on particular features.

Early Stopping - stops training when validation loss start increasing.

iii) Effect on generalization:

Overfitting means the model learns the training data too closely, including noise. Therefore, although training performance improves, performance on unseen data decreases.

5- Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs.

a) Epoch 1: $TA = 60\%$, $VA = 58\%$.

b) Epoch 2: $TA = 65\%$, $VA = 63\%$.

c) Epoch 3: $TA = 70\%$, $VA = 68\%$.

d) Epoch 4: $TA = 75\%$, $VA = 72\%$.

e) Epoch 5: $TA = 80\%$, $VA = 73\%$.

f) Epoch 6: $TA = 85\%$, $VA = 72\%$.

g) Epoch 7: $TA = 88\%$, $VA = 70\%$.

h) Epoch 8: $TA = 92\%$, $VA = 68\%$.

i) When does overfitting start?

Overfitting starts at epoch 6.

Validation accuracy reaches its highest value of 75% at epoch 5 and then starts decreasing, while training accuracy continues increasing.

ii) Why does validation accuracy decrease?

The model starts memorizing the training data instead of learning general patterns. Therefore:

→ Training accuracy keeps increasing.

→ Validation accuracy decreases.

→ The model performs poorly on unseen data.

iii) Two methods to improve validation performance

Dropout - reduces over-dependence on specific neurons.

Early Stopping - stops training around the point where validation performance is best.